Project Report

ELEC 5705

Behavioral Reimplementation of a
Background-Calibrated 13-bit SAR ADC

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Abstract

This project reimplements the background calibration architecture of the 46 μW, 13-bit, 6.4 MS/s SAR ADC reported by Ding *et al.* using a hybrid methodology. The ADC core, DAC mismatch, comparator modes, and analog correction path are modeled behaviorally, while the sign extraction and immediate calibration-control path are implemented explicitly in RTL. The final model reproduces sparse background operation and converges in 384243 conversions with an active calibration rate of approximately 1.37%, close to the paper's reported convergence scale of roughly 400k cycles.

1 Objective

The paper [1] uses redundancy and a sign-based background calibration loop to suppress DAC mismatch and comparator dynamic offset while keeping the ADC power very low. The project target was to reimplement that calibration architecture without going transistor-level. Following the project guidance, the implementation split used here is:

- behavioral modeling for the ADC core and analog correction plant, and
- explicit RTL for the sign-extraction path and the immediate calibration-control logic.

This yields a model that still captures the paper's [1] central contribution: the calibration loop only needs the *sign* of the error, not a full digital estimation engine.

In practical terms, the project focuses effort where the calibration architecture is most distinctive. The converter front end is kept at the behavioral level so that mismatch, redundancy, and comparator-mode effects can be exercised over long simulations, while the calibration decision path is forced to exist as real logic that captures trigger interpretation, sign extraction, filtering, and trim update.

2 Architecture

Figure 1 shows the overall project partition. The behavioral side produces the internal raw code and the modeled comparator/DAC behavior. The explicit RTL side interprets calibration triggers, extracts the sign from D_{15} and D_{16}, filters the sign decisions through a 6-bit bidirectional counter, and updates the trim registers.

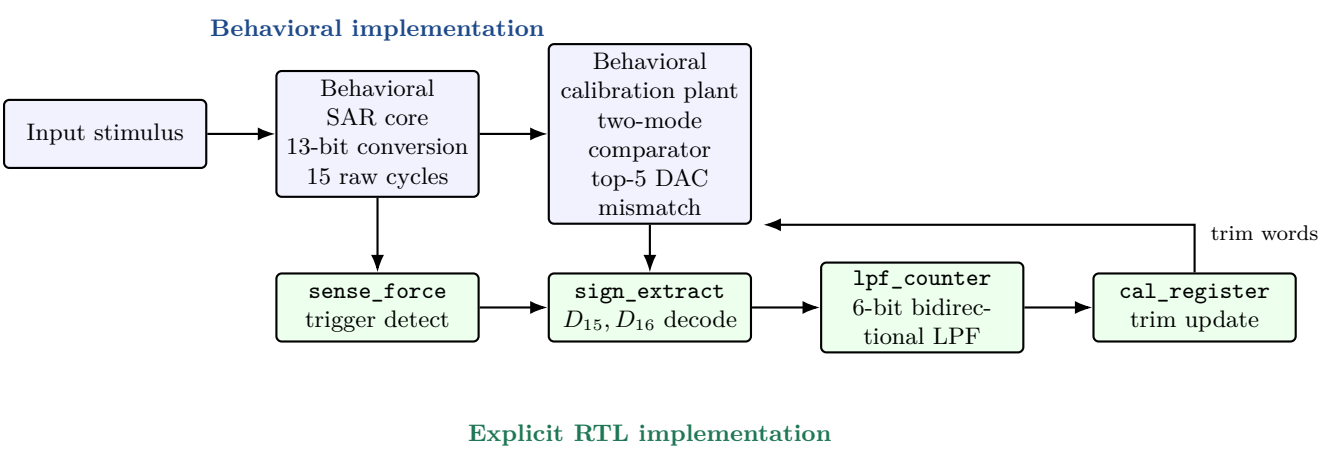


Figure 1: Hybrid project architecture. The converter and correction plant remain behavioral, while the sign-extraction and trim-update path are explicitly implemented in RTL.

Figure 2 shows the implemented sign path in a paper-style cycle-15/cycle-16 view based on the calibration method in [1]. For comparator calibration, the DAC code is held fixed while the comparator mode changes between the two final cycles. For DAC calibration, cycle 16 forces the alternate redundant A/B code. In both cases, the sign logic only examines whether D_{15} and D_{16} differ and converts that difference into the correction direction required by the trim hardware.

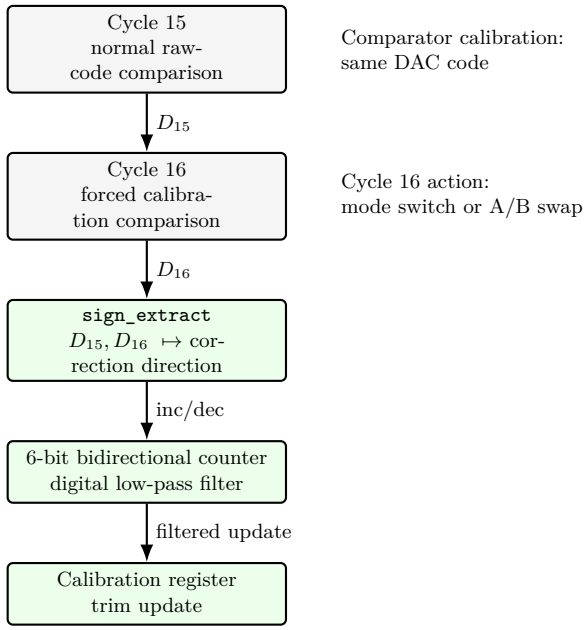


Figure 2: Implemented sign-extraction path. The logic uses the final two comparison outcomes to determine correction direction, then applies a 6-bit bidirectional counter as the digital LPF described in the paper.

3 Implemented Blocks

3.1 Sense and Force

The `sense_force` block is implemented as a purely combinational decoder. It receives the 15-bit behavioral raw code and splits it into a 7-bit `x_code` field and a 3-bit `y_code` field. Calibration is permitted only when `y_code == 3'b110`, which reproduces the centered background-calibration activation idea described in [1]. Once that gate is met, the block checks the `x_code` against one comparator family and five DAC A/B code pairs.

Comparator calibration is selected when `x_code[6:2] == 5'b11000`. In that case the block asserts `cal_active`, sets `cal_mode_comp = 1`, routes the request to channel 5, and forwards `x_code[1]` as `ab_sel`. DAC calibration uses ten explicit 7-bit codes, two per channel, covering the top five capacitors. For example, channel 0 uses `0111111/1000000`, channel 1 uses `0101111/0110000`, and the remaining channels continue inward toward midscale in the same A/B-pair form. The outputs of this block are therefore not abstract labels; they are the actual control signals used by the rest of the implemented calibration chain.

3.2 Sign Extraction

The sign-extraction block is the main non-behavioral requirement of the project, and it is implemented as combinational logic. The block takes five inputs: `valid`, `cal_mode_comp`, `ab_sel`, `d15`,

and `d16`. In the current implementation the decision rule itself is intentionally compact: if `valid` is low, or if `d15` and `d16` are equal, the block emits no sign information. Only when `valid && (d15 ^ d16)` is true does it declare a useful sign event.

Within that valid event window, the block computes

$$\texttt{sign_pos} = D_{15}\overline{D_{16}}, \qquad \texttt{sign_neg} = \overline{D_{15}}D_{16}.$$

The outputs `inc_i` and `dec_i` are then driven directly from `sign_pos` and `sign_neg`. In other words, the implementation does not calculate an error magnitude and then quantize it. It simply latches the relative ordering of the final two comparison results and converts that ordering into a one-cycle increment or decrement request for the trim loop. The sign convention is therefore tied to the correction direction required by the implemented trim hardware rather than to an abstract algebraic sign.

Table 1: Implemented sign-extraction mapping.

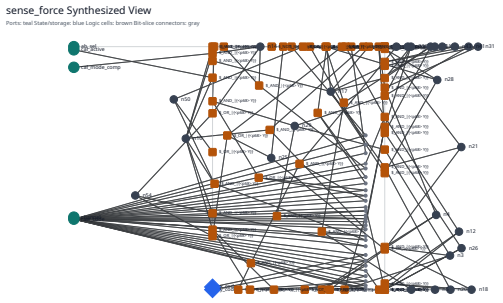
Valid	D_{15}	D_{16}	sign_valid	inc_i	dec_i
0	X	X	0	0	0
1	0	0	0	0	0
1	1	1	0	0	0
1	1	0	1	1	0
1	0	1	1	0	1

3.3 Digital LPF and Trim Update

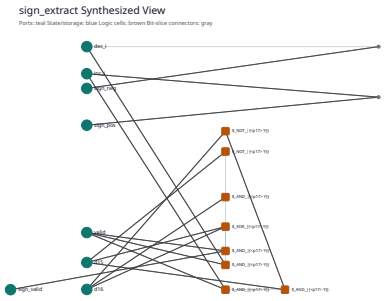
The paper's [1] digital low-pass filter is implemented as a 6-bit bidirectional counter, and the project follows that structure directly. Each counter is clocked and resettable, starts at `INIT = 32`, and ignores contradictory requests. When `inc_i` is asserted by itself, the count moves upward by one. When `dec_i` is asserted by itself, the count moves downward by one. If the counter reaches one count before either edge, it does not remain saturated; instead it emits a one-cycle filtered update pulse (`inc_o` or `dec_o`) and immediately reloads to its midpoint. This reproduces the intended effect of requiring many repeated sign decisions before an actual trim change is allowed.

The trim word update is implemented separately as a synchronous saturating register with a 7-bit trim word. On `inc_o` it increments by one unless already at all ones, and on `dec_o` it decrements by one unless already at zero. In the full project simulation there are six copies of the LPF-plus-register path: five for the DAC capacitor channels and one for the comparator channel. This means the implemented digital portion of the project is not only the sign extractor itself, but the complete chain from trigger detection through filtered trim update.

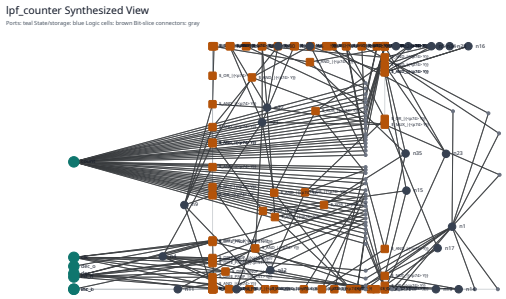
In addition to functional simulation, the explicitly implemented digital calibration path was synthesized as a standalone RTL block. This synthesis excludes the behavioral SAR core and analog plant, and keeps only the logic that was actually implemented: trigger decoding, sign extraction, six LPF counters, and six trim registers. Figure 3 shows the separately synthesized views of the four unique RTL blocks used in that path.



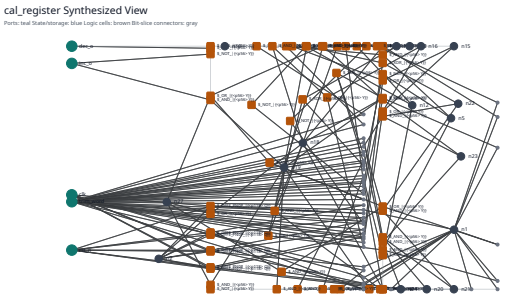
(a) sense_force



(b) sign_extract



(c) lpf_counter



(d) cal_register

Figure 3: Separately synthesized views of the four implemented RTL blocks that form the digital calibration path.

At the hierarchy level, the synthesized top block contains one `sense_force`, one `sign_extract`, six `lpf_counter` instances, and six `cal_register` instances. After flattening, the complete implemented digital path maps to 941 primitive cells, including 90 sequential elements and 851 combinational logic cells.

Table 2: Synthesized digital calibration-path summary.

Metric	Result
Top-level hierarchy	1 `sense_force`, 1 `sign_extract`, 6 `lpf_counter`, 6 `cal_register`
Flattened total cells	941
AND gates	419
OR gates	202
NOT gates	127
XOR gates	91
MUX cells	12
Sequential cells	90

4 Behavioral Model

The converter core and analog correction path remain behavioral. This includes the 13-bit conversion source, the simplified 15-bit redundant raw-code encoding, the two-mode comparator offset

plant, the top-five DAC mismatch model, and the analog effect of the trim words. Within this model, D_{15} and D_{16} are generated from actual behavioral DAC/comparator comparisons, so the implemented sign-extraction logic operates on comparison outcomes that are consistent with the calibration mechanism described in [1].

The DAC side uses explicit trim-dependent weight perturbations for the top five capacitors, so a change in trim word directly changes the converter transfer behavior. The comparator side uses two effective operating modes together with a residue-dependent dynamic offset term, which allows the calibration loop to act on the kind of signal-dependent comparator error discussed in the paper [1] rather than on a single static threshold shift. This is still a system-level model, but it is much closer to the implemented calibration mechanism than the earlier placeholder-error version.

5 Results

5.1 Single-Loop Sanity Checks

Before enabling the full six-channel background loop, simplified single-loop testbenches were used to verify that the explicit sign-extraction path drives the trim in the correct direction. Figure 4 shows one comparator-offset case and one DAC case.

These sanity-check plots serve as a digital-path validation step. They confirm that the sign-extraction output, the bidirectional counter, and the trim register all agree on the required correction direction before sparse trigger activation is introduced.

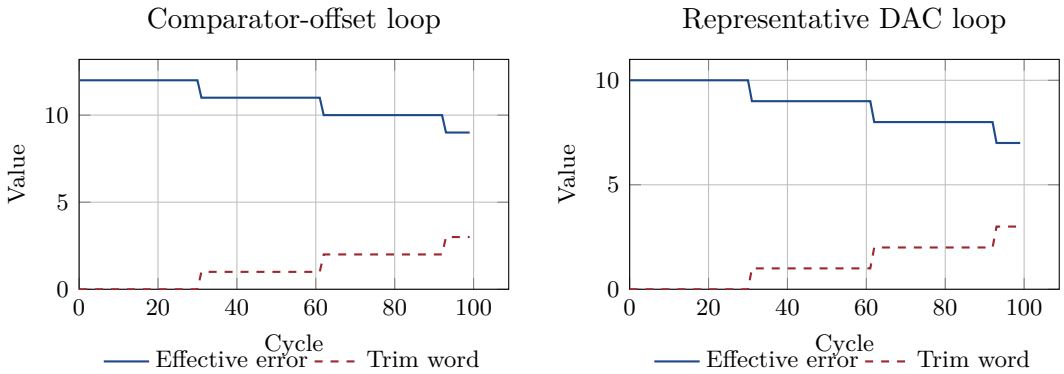


Figure 4: Single-loop sanity checks used to verify that the implemented sign path moves the trim word in the expected direction.

5.2 Background Calibration Convergence

The full background-calibration experiment includes one comparator channel and five DAC-capacitor channels. Figure 5 plots the six trim words as the background calibration proceeds. These traces show internal register evolution rather than direct ADC performance, so they indicate when each calibration channel reaches its final stored correction code. The long settling time is a direct result of sparse trigger-code activation rather than continuous correction on every conversion.

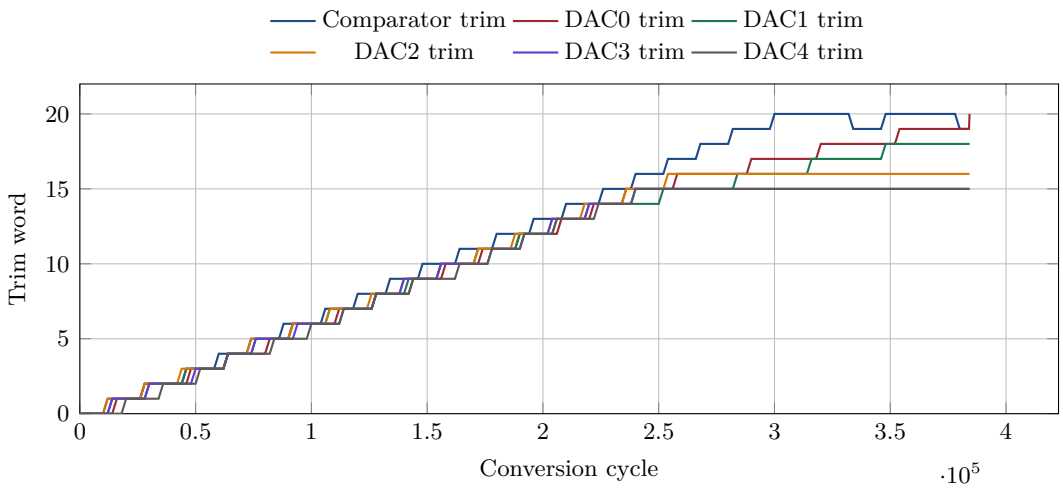


Figure 5: Convergence of the six calibration trim words in the full background run.

To mirror the form of the paper's Fig. 22 [1], Figure 6 plots the system-level SNDR obtained by evaluating the implemented DAC-mismatch equations and the implemented two-mode comparator model at the trim states recorded during the background-calibration run. This figure therefore represents ADC performance convergence, not the exact moment when every internal trim register reaches its final code. The plotted curve is generated directly from those sampled states and then smoothed for presentation so the long-term convergence trend is easier to read. After convergence, the evaluated SNDR reaches approximately 73.9 dB.

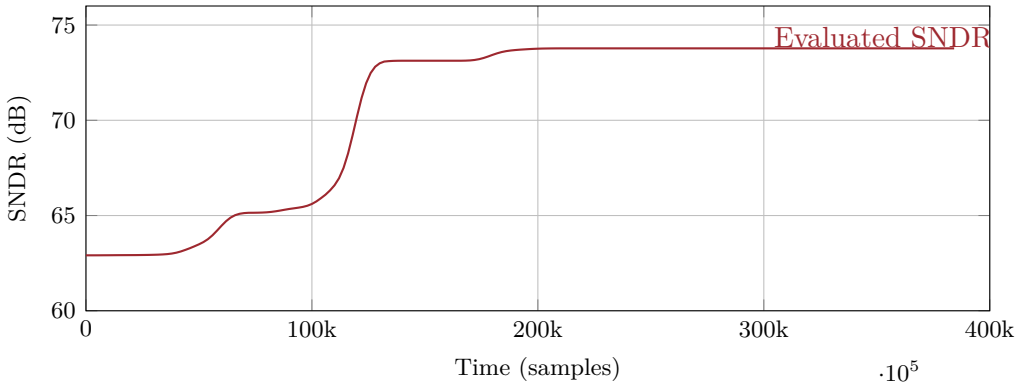


Figure 6: SNDR-convergence plot in the style of the paper's Fig. 22, evaluated directly from the implemented dynamic SAR output model using the recorded trim trajectory. The plotted curve is a smoothed rendering of the sampled SNDR values, and the peak converged SNDR is approximately 73.9 dB.

Figure 7 recreates the paper's [1] three-scenario spectrum comparison using the same implemented dynamic SAR output model: without calibration, with comparator-offset calibration only, and with a fully corrected comparator-plus-DAC trim state. Figure 6 shows how the calibrated SNDR develops over the logged background-calibration run, while Figure 7 compares three representative operating points from that same model. Taken together, they show both the convergence process and the final spectral improvement. In these results, comparator calibration removes the dominant mode-dependent comparator distortion, and the calibrated DAC weights contribute a further improvement of roughly 3.1 dB in SNDR. Because SNDR is a system-level performance

metric, it approaches its final value before every individual trim channel reaches its exact final stored code.

Using the standard relation

$$\mathrm{ENOB} = \frac{\mathrm{SNDR} - 1.76}{6.02},$$

the three spectral cases correspond to approximately 10.2 bits without calibration, 11.5 bits with comparator calibration only, and 12.0 bits with both comparator and DAC calibration enabled. This ENOB view is useful because it expresses the same spectral improvement in directly ADC-oriented terms: the implemented calibration loop recovers about 1.8 effective bits relative to the uncalibrated case, while the DAC calibration path still adds roughly 0.5 effective bits beyond comparator-only correction.

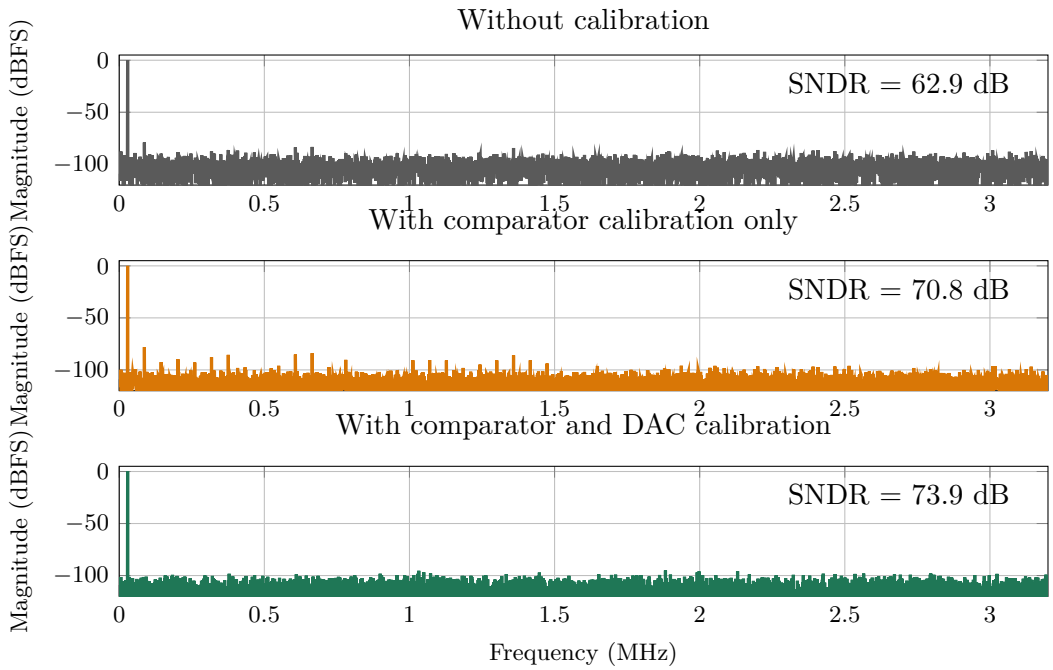


Figure 7: Spectrum comparison in the style of the paper's Fig. 19, evaluated from the implemented dynamic SAR output model for three calibration states: without calibration, with comparator calibration only, and with a fully corrected comparator-plus-DAC state.

The main convergence metrics from the background-calibration run are summarized in Table 3. These numbers should be read as internal channel-settling times: they report when each trim register first reaches its final stored correction value, not when the overall ADC SNDR first appears visually settled in Figure 6.

Table 3: Background calibration convergence summary.

Metric	Result
Total convergence cycle	384243
Active calibration events	5257
Active-event rate	1.37%
Comparator loop convergence	299008
DAC0 convergence	384243
DAC1 convergence	346060
DAC2 convergence	252461
DAC3 convergence	219268
DAC4 convergence	190316

6 Discussion

The most important result is that the implemented calibration architecture operates in the same regime as the paper [1]. The six trim loops converge over hundreds of thousands of conversions rather than over a short foreground-like interval, and the activation rate remains low at approximately 1.37%. This is consistent with the intended background nature of the calibration.

The results also separate the roles of the two calibration categories. Comparator calibration produces the larger initial gain because the comparator term dominates the uncalibrated distortion in the present model. DAC calibration still matters: once the comparator term is corrected, the five DAC trim channels provide a further SNDR increase and improve ENOB from roughly 11.5 bits to roughly 12.0 bits. This behavior is consistent with a system in which comparator error is the first-order limitation and DAC mismatch is the next-order residual limitation.

The model is still intentionally behavioral. The trigger set is paper-faithful in structure but not a literal transistor-level reconstruction of every published detail in [1], and the analog trim effect is represented with a compact weight-perturbation model. That simplification is deliberate: the project emphasis is on the calibration architecture, especially the explicitly implemented sign-extraction path and its interaction with the LPF and trim registers, rather than on full circuit-level replication.

7 Conclusion

This project reimplemented the key background-calibration architecture of the 13-bit SAR ADC reported by Ding *et al.* [1] using the hybrid split required for the course project. The SAR conversion path, DAC mismatch, comparator operating modes, and analog trim action were modeled behaviorally, while the trigger decoding, sign extraction, 6-bit bidirectional filtering, and trim-word update path were implemented directly in RTL. In that sense, the project does not simply describe the calibration algorithm; it implements the digital decision path that makes the sign-based background calibration work.

The resulting system reproduces the main behaviors emphasized by the paper. The background loop converges in 384243 conversions with an activation rate of 1.37%, which is close to the paper's reported convergence scale. The same implemented model also shows a progression from about 62.9 dB SNDR without calibration to about 70.8 dB with comparator calibration and about 73.9 dB with both comparator and DAC calibration. Expressed as ENOB, this corresponds to an improvement

from about 10.2 bits to about 12.0 bits.

Overall, the project demonstrates that the sign-based background calibration architecture can be reimplemented faithfully at the behavioral-plus-RTL level without going transistor-level. The explicit RTL path captures the core calibration logic, and the behavioral plant is detailed enough to reproduce sparse convergence, separate comparator and DAC correction effects, and produce results that are meaningfully comparable to those reported in [1].

References

[1] M. Ding, P.-I. Mak, M.-K. Law, S.-W. Sin, and R. P. Martins, "A 46 ţW 13-b 6.4-MS/s SAR ADC with background mismatch and offset calibration," *IEEE Journal of Solid-State Circuits*, vol. 52, no. 10, pp. 2466–2477, Oct. 2017.